

Internal Unified Theory Note: Damping-Operator Hierarchy, Rational Self-Energy, and Flow-Feasible Retrofitting in IBR Grids

Project working document

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Abstract

This document is the internal map of the project. It is not a submission draft. It organizes the verified results, failed hypotheses, and open gates under one operator hierarchy. The central thesis is that the transition from synchronous grids to inverter-rich grids is a descent through a damping-operator hierarchy: proportional viscous damping, non-proportional viscous damping, and rational frequency-dependent damping induced by controller self-energy. Each descent breaks a classical guarantee and creates a planning risk. Level 0 gives normal modes and topology-invariant decay under proportional damping. Level 1 breaks classical modal separation; heterogeneous D/M creates modal coupling, dynamic Braess-like reversals, and non-monotone inertia effects. Level 2 condenses controller states into a rational self-energy

$$\Sigma(s) = D_0 + C(sI - A_{cc})^{-1}B,$$

which yields a nonlinear eigenvalue problem. This bridge was validated against full ANDES eigensolves on IEEE 39-bus cases, including a 13.7417 Hz PLL-family critical mode in Mix60. The document also integrates the newer resolvent-margin and coordinated-retrofitting gates. Those gates support a secondary but useful claim: modal damping-ratio and protected-resolvent certificates rank actions differently across several protection weightings, and equal-cost packages $w + D$ and $w + M$ often beat the best single lever under the protected resolvent. The mechanism of that certificate discrepancy is still open; simple global non-normality and commutator descriptors correlated weakly. Every claim below is labeled by validation status.

Contents

1	Executive Thesis	3
1.1	The operator hierarchy	3
1.2	The material-active candidate framing	3
1.3	What papers can come out	4
2	Validation Ledger	4
3	Level 0: The Classical Floor	6
4	Level 1: Non-Proportional Viscous Damping	7
4.1	QEP sensitivities	7
4.2	Why inertia can be deceptive	8
4.3	The line Braess dichotomy	8
4.4	Second-order modal correction	9

5	Level 2: Rational Nonviscous Bridge	9
5.1	Schur self-energy	9
5.2	Beyn contour validation	10
5.3	Active nonviscous material	10
6	Resonant Braess and Hidden Control Margin	10
7	Protected Resolvent Certificates and Coordination	11
7.1	Two margins	11
7.2	Bi-orthogonal resolvent sensitivity	12
7.3	Gate results	13
7.4	Flow-feasible resolvent retrofitting algorithm	13
8	Weak Nodes, Weak Links, and Substitutability	13
9	Open Gates	14
10	What Is Actually New	15
11	Submission Map	15

1 Executive Thesis

The project can be read as one question:

What object preserves damping-margin mechanisms after IBR controls are condensed?

The answer changed as the work progressed. A scalar (L, M, D) bridge is useful for intuition but not enough for detailed IBR dynamics. The correct reduced object for the ANDES cases is a rational Schur self-energy or, equivalently, a retained-state nonlinear eigenvalue problem (NEP). Once that bridge exists, we can ask smaller planning questions: which line is weak, which node is weak, which damping/inertia/topology action moves the margin, and whether packages of actions beat single levers under flow feasibility.

1.1 The operator hierarchy

The useful organizing device is a three-level hierarchy of damping operators. Each level has a mathematical owner in the literature, a guarantee, and a planning risk when the grid leaves that level.

Level	Operator	Classical guarantee	Planning risk when violated
0	Proportional viscous damping, $D = \gamma M$ or Rayleigh/Caughey-class damping	Normal-mode separation; common decay for underdamped proportional modes; topology moves frequency more than decay	Frequency-only planning can look valid, but only inside the proportional floor
1	Constant non-proportional viscous damping, $[\tilde{L}, \tilde{D}] \neq 0$	Modal coordinates are no longer decoupled; off-diagonal modal damping moves poles	Line reinforcement or inertia can reduce damping ratio; Fiedler/frequency rankings can fail
2	Rational/nonviscous damping, $\Sigma(s) = D_0 + C(sI - A_{cc})^{-1}B$	Constant-coefficient QEP is no longer the reduced object; controller poles can become critical	Static bridge and fixed-point screens can miss control-family critical modes

This hierarchy is not a claim that Rayleigh damping, non-proportional modal perturbation, nonviscous damping, Schur complements, or contour eigensolvers are new. They are not. The project contribution, if it survives adversarial review, is the power-system formulation and validation: treating IBR grids as active nonviscous networked structures and using a controller-aware rational bridge to classify network-family and control-family margin mechanisms.

1.2 The material-active candidate framing

The most compact physical sentence is:

An IBR-dominated grid behaves like a viscoelastic structure made of active material: its controller self-energy can have oscillatory, band-active poles, so nonviscous modes can become the critical small-signal modes.

This is a candidate framing, not a closed novelty claim. Passive viscoelastic models have a long literature, including hereditary kernels and nonviscous damping models. The power-system observation is that a PLL or inverter-control kernel is not a passive relaxation kernel in the same sense. In Mix60, the critical pole is a 13.7417 Hz control-family mode with $\pi_c = 0.853$, close to a PLL-dominated condensed-control pole. That is the empirical anchor. The adversarial literature search for “active viscoelastic structures,” active nonviscous kernels, and nonviscous damping in power grids remains open.

1.3 What papers can come out

- **Paper 1: the controller-aware bridge.** This is the strongest validated paper. It reports the scalar bridge failure, the Schur NEP bridge, Beyn contour extraction, network/control-family classification, Mix60’s control-family critical mode, and the resonant Braess mechanism with negative control.
- **Paper 2: the hierarchy and passivity gate.** This is conditional. It becomes strong if a passivity/active-kernel test separates cases where the static bridge is safe from cases where Level 2 is required.
- **Paper 3: flow-feasible resolvent retrofitting.** This is a conference-scale candidate. The Gate 1 and Gamma gates show a real phenomenon: protected-resolvent and modal certificates recommend different actions, and $w + D$ or $w + M$ packages often beat single levers under the protected resolvent. The missing piece is a convincing local mechanism descriptor.
- **Thesis material.** The weak-node/weak-link maps, substitutability spectrum, inertia reversals, and graph-configuration optimization are useful thesis chapters. Several are formulated or surrogate-verified rather than full ANDES-validated.

2 Validation Ledger

Table 1 is the most important table in the document. It separates validated bridge results from formulated planning ideas and dead ends.

Table 1: Claim ledger. Status terms are intentionally blunt.

Claim	Evidence	Control or comparison	Status
Scalar (L, M, D) bridge is insufficient for detailed ANDES damping	In IEEE39 no-PSS/base/Mix60, local scalar damping from GENROU.D is zero while modal damping exists; Phase-0D reports scalar D norm zero and nonzero Schur damping diagnostics.	Full ANDES eigensolve and Schur diagnostics.	ANDES-validated failure mode
Controller-aware Schur NEP bridge recovers critical poles	Phase-0D: no-PSS, base, Mix60 pass. Worst reported errors: Mix60 frequency $8.475 \times 10^{-11}\%$, damping $2.479 \times 10^{-9}\%$.	Full ANDES eigensolve; contour argument counts; non-circular polish.	ANDES-validated
Mix60 critical mode is control-family	Pole $-10.7514 + j86.3419$, $f = 13.7417$ Hz, $\zeta = 0.12357$, $\pi_c = 0.853$; nearest A_{cc} pole distance 0.215; top reconstructed states PLL1/REGF1.	Scalar bridge and fixed-point Schur reduction missed or misidentified the mode.	ANDES-validated mechanism anchor
Resonant Braess mechanism exists in benchmark	Phase Braess-NEP: 40/348 strict resonant records. Clearest case Line 42 (20–34), self-energy term C dominates with control denominator fraction 0.9594 and median analytic-vs-NEP smooth error 8.08×10^{-4} .	Requires network-family damping decrease, closer A_{cc} pole, PLL-dominated nearest pole, and dominant self-energy term; soft-GFL negative control removes resonances.	ANDES/NEP mechanism validated, minority and local
Second-order modal correction improves diagonal screen in weak/mild regimes	Synthetic and reduced-PLL audits: weak surrogate improvement from diagonal to second order; reduced PLL reported $2.16\% \rightarrow 0.09\%$ median in prior cross-check.	Diagonal screen and reduced QEP fallback.	Numerically verified outside full ANDES; estimator validation pending

Claim	Evidence	Control or comparison	Status
Second-order correction always beats large reduced QEP	Refuted by cross-checks; QEP $r = 20$ can beat or match second order in tails.	Reduced QEP size sweep.	KILLED
Inertia can reduce damping margin	Fixed- D_i/M_i surrogate audit: 5447/16229 perturbations reduce both diagonal and full-QEP margin. ANDES GENROU.M gate: no-PSS/base have 5/10 negative GENROU.M perturbations.	Fixed- D_i versus fixed- D_i/M_i control; full ANDES eigensolve for GENROU.M cases.	Valid for mechanical inertia checks; virtual inertia GFM blocked
Virtual-inertia assignment works over controller-aware IBR bridge	REGF1 sheets expose no clear M , H , or T_j knob. Mix60 response to retained GENROU.M is weak signal.	Phase inertia bridge gate.	BLOCKED; do not claim
Certificate discrepancy S_ζ vs S_g	Gate 1: ranking agreement 28.3%. Gamma gate: physical profiles have agreement 21–29%.	Four physical Gamma choices and eight random diagonal profiles.	Robust phenomenon, mechanism weak
Coordinated packages beat single levers at equal cost	Gate 1/Gamma: under S_g , $w + D$ has $CV < 0$ in 72.1–73.8%, $w + M$ in 67.2–73.8%; $D + M$ not robust.	Equal-cost budget, DC flow feasibility, same seeds, Gamma profiles.	Reduced-model gate supports $w + D$ and $w + M$ only
Nonlinear synergy of levers	SynNorm < -0.10 never exceeds 50% in Gate 1.	Three eps sizes and all lever pairs.	KILLED as headline
Cross inertia-control term is empirically frequent in ANDES	Phase E0 found 0/13 measurable singleton cases.	ANDES bridge checks.	KILLED as empirical headline; remains local theory

3 Level 0: The Classical Floor

The starting point is the second-order model

$$M\ddot{x} + D\dot{x} + Lx = 0, \quad (1)$$

where $M \succ 0$ and $L = L^T \succeq 0$. Inertia-normalized coordinates $y = M^{1/2}x$ give

$$\ddot{y} + \tilde{D}\dot{y} + \tilde{L}y = 0, \quad \tilde{L} = M^{-1/2}LM^{-1/2}, \quad \tilde{D} = M^{-1/2}DM^{-1/2}. \quad (2)$$

If $D = \gamma M$, then $\tilde{D} = \gamma I$ and every eigenvector of $\tilde{L}$ is also an eigenvector of $\tilde{D}$. Writing $\tilde{L}q_k = \mu_k q_k$, each nonzero mode obeys

$$\ddot{z}_k + \gamma \dot{z}_k + \mu_k z_k = 0, \quad (3)$$

with poles

$$s_{k,\pm} = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\mu_k}}{2}. \quad (4)$$

For underdamped modes, $\mu_k > \gamma^2/4$, the real part is exactly

$$\text{Re}(s_{k,\pm}) = -\gamma/2. \quad (5)$$

Proposition 1 (Topology-invariant decay at the proportional floor) *Under $D = \gamma M$, every mode that remains underdamped has decay rate $-\gamma/2$, independent of topology.*

This is a floor theorem, not a project contribution. It is the same normal-mode logic behind classical proportional damping and Rayleigh damping. It explains why many graph-frequency intuitions feel safe in homogeneous systems. The escape channel is also clear: topology can move μ_k across the overdamped/underdamped boundary. The rigorous statement is not that topology never matters. It is that decay is invariant for modes that stay in the underdamped proportional class.

The project validation notes recorded numerical deviations at the level of 5.8×10^{-16} to 1.4×10^{-16} in proportional tests. Those numbers are useful sanity checks, not new theory.

4 Level 1: Non-Proportional Viscous Damping

Level 1 begins when $\tilde{D}$ and $\tilde{L}$ no longer share eigenvectors. In modal coordinates of $\tilde{L}$,

$$\ddot{z} + \Gamma \dot{z} + \Lambda z = 0, \quad \Gamma = Q^T \tilde{D} Q, \quad \Lambda = \text{diag}(\nu_k). \quad (6)$$

Write

$$\Gamma = \Delta + E, \quad \Delta = \text{diag}(\delta_k), \quad E_{kk} = 0. \quad (7)$$

The diagonal screen uses scalar modal polynomials

$$p_k(s) = s^2 + \delta_k s + \nu_k. \quad (8)$$

The actual QEP is coupled by sE . This is the first place where topology, damping, and inertia can interact in non-intuitive ways.

4.1 QEP sensitivities

For the QEP

$$P(s)v = (s^2 M + sD + L)v = 0, \quad (9)$$

assume a simple eigenvalue and use the complex-symmetric bilinear product $v^T(\cdot)v$ when M, D, L are real symmetric. Define

$$\xi = v^T(2sM + D)v. \quad (10)$$

Then the local parameter sensitivities are

$$\frac{\partial s}{\partial M_i} = -\frac{s^2 v_i^2}{\xi}, \quad (11)$$

$$\frac{\partial s}{\partial D_i} = -\frac{s v_i^2}{\xi}, \quad (12)$$

$$\frac{\partial s}{\partial w_e} = -\frac{(b_e^T v)^2}{\xi}, \quad (13)$$

where b_e is the signed incidence vector of line e in the selected coordinates. These formulas are standard perturbation theory. In the project they were used as a common local currency for topology, damping, and inertia. They were checked against central differences with relative errors around 10^{-8} in the three channels in the associated validation notes.

4.2 Why inertia can be deceptive

At the proportional point $D = \gamma M$, holding D fixed while perturbing M_i breaks proportionality. That direction is not the same as increasing both M_i and D_i proportionally. For an underdamped proportional mode, the directional derivative of the real part can move the pole right; in the project notes the exact local rate was written as a positive proportional quantity of the modal participation. The correct interpretation is damping dilution: if damping is held fixed and inertia is added, the ratio between energy storage and dissipation changes.

The stronger phrase “inertia is universally Braessian” is false and must not be used. Monte Carlo and fixed-ratio controls rejected it. The defensible statement is narrower:

Under heterogeneous damping, increasing inertia frequently but not universally reduces damping margin; at the proportional point, the harmful direction can be computed exactly when damping is held fixed.

The current full-ANDES evidence is also limited. In no-PSS and base IEEE39 checks, retained mechanical GENROU.M perturbations had 5/10 negative margin effects and the local NEP sensitivity matched full ANDES. In Mix60, the critical mode was control-family and the retained GENROU.M signal was too weak. No GFM virtual-inertia claim is available until a physical REGF1 inertia knob is identified.

4.3 The line Braess dichotomy

At Level 0, small-signal Braess-like damping degradation is blocked for underdamped proportional modes because line actions move modal frequency but not decay. At Level 1, off-diagonal modal damping allows a line action to move both the frequency and the decay rate. The damping-ratio derivative is

$$\frac{\partial \zeta}{\partial \rho} = -\frac{1}{|s|} \operatorname{Re} \left(\frac{\partial s}{\partial \rho} \right) + \frac{\operatorname{Re}(s)}{|s|^3} \operatorname{Re} \left(\bar{s} \frac{\partial s}{\partial \rho} \right), \quad (14)$$

so a line can improve a graph stiffness proxy while reducing damping ratio. The project found such reversals in surrogate and NEP audits. The resonant Level-2 version is stronger and appears below.

4.4 Second-order modal correction

Let s_0 be a diagonal-screen pole of the critical mode c :

$$p_c(s_0) = s_0^2 + \delta_c s_0 + \nu_c = 0. \quad (15)$$

To second order in E , eliminating the noncritical modal coordinates gives

$$\Delta s_c = -\frac{s_0^2}{2s_0 + \delta_c} \sum_{\ell \neq c} \frac{E_{c\ell} E_{\ell c}}{s_0^2 + \delta_\ell s_0 + \nu_\ell} + O(\|E\|^3), \quad (16)$$

up to sign convention depending on whether the eliminated term is moved to the left or right side. The useful object is the denominator:

$$H_{\ell\ell}(s_0) = \frac{1}{s_0^2 + \delta_\ell s_0 + \nu_\ell}. \quad (17)$$

Then the correction is a rational graph/modal filter:

$$\Delta s_c \propto [E H(\Lambda, \Delta; s_0) E]_{cc}. \quad (18)$$

This is not a new perturbation theorem. Its value in the project is interpretive: each term says which noncritical mode couples into the critical margin and how strongly its modal denominator amplifies the effect.

The empirical status is mixed in a useful way. The correction improves the diagonal screen in weak-to-mild non-proportional regimes and in a reduced PLL model. It does not replace a large reduced QEP in clustered or strongly coupled tails. Therefore the estimator is a workflow:

diagonal screen $\rightarrow$ second-order correction $\rightarrow$ reduced/full QEP or NEP fallback.

5 Level 2: Rational Nonviscous Bridge

5.1 Schur self-energy

Let the full linearized ANDES state matrix be partitioned as

$$A = \begin{bmatrix} A_{ee} & A_{ec} \\ A_{ce} & A_{cc} \end{bmatrix}, \quad (19)$$

where x_e are retained network/interface states and x_c are condensed controller, exciter, stabilizer, PLL, governor, and auxiliary states. The Schur NEP is

$$S_e(s) = sI - A_{ee} - A_{ec}(sI - A_{cc})^{-1}A_{ce}. \quad (20)$$

When the retained coordinates are reduced to a second-order form, the same structure can be written as

$$T(s) = s^2 M + s \Sigma(s) + \tilde{L}, \quad \Sigma(s) = D_0 + C(sI - A_{cc})^{-1}B. \quad (21)$$

This is the Level-2 object. It is a rational nonviscous damping operator. It is also the correct place to connect power-system IBR controls to the structural-dynamics literature on hereditary and nonviscous damping.

5.2 Beyn contour validation

The Phase-0D gate solved (20) by Beyn contour integration. No fixed-point iteration was used for acceptance. The contour solver first extracted zeros from contour moments and only afterward compared them with ANDES poles.

Case	retained	condensed	pole family	f Hz	ζ	critical error
no-PSS	20	110	network	1.3705	0.15447	$4.861 \times 10^{-14}\%$ freq
base	20	140	network	1.3706	0.15443	$1.62 \times 10^{-13}\%$ freq
Mix60	11	167	control	13.7417	0.12357	$8.475 \times 10^{-11}\%$ freq

For Mix60, the matched pole was

$$s_{\text{crit}} = -10.7514 + j86.3419, \quad (22)$$

with $\pi_c = 0.853$. The nearest condensed-control pole was at distance 0.215 and the reconstructed condensed states were dominated by PLL1 and REGF1 components. The scalar bridge and fixed-point reduction did not capture this mode. This is the strongest full-ANDES result in the project.

5.3 Active nonviscous material

Passive nonviscous damping kernels in structural dynamics are usually designed to dissipate energy. Their extra nonviscous modes are not expected to become the least-damped oscillatory modes in the same way a PLL-family mode can in an IBR grid. In contrast, the condensed IBR/control self-energy can have poles and frequency bands that act like active material. A concise way to write the candidate test is to define a passivity defect:

$$\Pi_{\text{pass}} = \max_{\omega \in \Omega} \left[-\lambda_{\min} \left(\frac{\Sigma(j\omega) + \Sigma(j\omega)^*}{2} \right) \right]_+. \quad (23)$$

The conjectured rule is:

Conjecture 1 (Passive-kernel screening safety) *If Π_{pass} is small on the relevant band and the Level-1 Caughey defect*

$$\chi_{\text{CK}} = \frac{\|\tilde{L}\tilde{D} - \tilde{D}\tilde{L}\|_F}{\|\tilde{L}\|_F \|\tilde{D}\|_F} \quad (24)$$

is small, then Level-0/1 screens are reliable for network-family modes. If Π_{pass} is large, control-family modes can become critical and the full rational bridge is required.

This is not yet a theorem. The Gamma gate already showed that simple global non-normality and global χ do not explain certificate discrepancy strongly. The passivity defect must therefore be tested locally at the critical mode and controller-pole band, not sold as an established result.

6 Resonant Braess and Hidden Control Margin

For a line reinforcement parameter ρ , the Schur NEP sensitivity is

$$\frac{\partial s}{\partial \rho} = -\frac{y^H S_\rho(s)x}{y^H S_s(s)x}, \quad (25)$$

where x and y are right and left null vectors of $S_e(s)$. The derivative $S_s(s)$ includes the resolvent-square term

$$A_{ec}(sI - A_{cc})^{-2}A_{ce}. \quad (26)$$

This term is the signature of frequency-dependent control self-energy.

The Phase Braess-NEP gate decomposed the damping-ratio movement into retained network terms and a condensed-control term. A reinforcement was called resonant only if it satisfied all of the following:

1. the tracked network-family damping ratio decreased;
2. the tracked mode moved closer to a matched A_{cc} pole;
3. the nearest condensed pole was PLL-dominated;
4. the self-energy term dominated the retained terms;
5. the effect survived finite-difference NEP/ANDES comparison.

The gate found 40 strict resonant records among 348 reversals. The clearest case was Mix60/no-PSS Line_42 (20–34), mode rank 2:

$$\zeta_{\text{margin}} : 0.123566 \rightarrow 0.123558, \quad \zeta_{\text{network}} : 0.133884 \rightarrow 0.133816,$$

with distance to the matched A_{cc} pole $3.33227 \rightarrow 3.3279$. The self-energy component was -6.874×10^{-3} , while the retained terms were orders of magnitude smaller, and the finite-difference derivative was -6.860×10^{-3} .

This result should be written as a mechanism result. It is not a claim that every line reinforcement or every IBR grid exhibits resonant Braess behavior. The soft-GFL negative control moved the system away from the resonance band and removed the strict resonant cases.

7 Protected Resolvent Certificates and Coordination

7.1 Two margins

The modal margin used throughout the earlier project is

$$S_\zeta = \log \frac{\zeta_{\text{req}}}{\zeta_{\text{crit}}}, \quad (27)$$

where lower is better and $S_\zeta \leq 0$ is safe. The protected-resolvent margin is

$$S_g = \log g_{\text{prot}}, \quad g_{\text{prot}} = \sup_{\omega \in \Omega} \sigma_1(\Gamma_y C(j\omega I - A)^{-1} B \Gamma_u). \quad (28)$$

The Gate 1 baseline used identity protection weights. The Gamma gate repeated the test with physically motivated and random diagonal protection weights.

7.2 Bi-orthogonal resolvent sensitivity

Let ω^* be a simple maximizer of (28), and let u^*, v^* be the leading left and right singular vectors of

$$H^* = \Gamma_y C(j\omega^* I - A)^{-1} B \Gamma_u.$$

Set

$$\phi^{\text{in}} = (j\omega^* I - A)^{-1} B \Gamma_u v^*, \quad (29)$$

$$\phi^{\text{out}} = (j\omega^* I - A)^{-H} C^H \Gamma_y^H u^*. \quad (30)$$

For a parameter p that changes A but not the protection weights, the local sensitivity is

$$\frac{\partial S_g}{\partial p} = \frac{1}{g_{\text{prot}}} \text{Re} \left[(\phi^{\text{out}})^H \frac{\partial A}{\partial p} \phi^{\text{in}} \right], \quad (31)$$

provided the peak and singular value are simple. This is standard singular-value perturbation applied to the resolvent. Its value is that all levers can be priced in the same margin units.

For the swing first-order matrix

$$A = \begin{bmatrix} 0 & I \\ -M^{-1}L & -M^{-1}D \end{bmatrix},$$

write $\phi = (\phi_\theta, \phi_\omega)$. Then local damping at bus i has

$$\frac{\partial A}{\partial D_i} = \begin{bmatrix} 0 & 0 \\ 0 & -M_i^{-1} e_i e_i^T \end{bmatrix}, \quad (32)$$

so

$$\frac{\partial S_g}{\partial D_i} = -\frac{1}{g_{\text{prot}} M_i} \text{Re} \left[\overline{\phi_{\omega,i}^{\text{out}}} \phi_{\omega,i}^{\text{in}} \right]. \quad (33)$$

This is the corrected version of the Gemini suggestion. It is not generally $-|\phi_i^{\text{out}}|^2$ unless the system is normal or the input and output spatial vectors align. The important point is the bi-orthogonal product. In a non-normal grid, the forcing vector and response vector can be spatially different. That is the mathematical source of spatial decoupling.

For a line $e = (i, j)$ with incidence b_e , L changes by $b_e b_e^T$, giving

$$\frac{\partial A}{\partial w_e} = \begin{bmatrix} 0 & 0 \\ -M^{-1} b_e b_e^T & 0 \end{bmatrix}, \quad (34)$$

and therefore

$$\frac{\partial S_g}{\partial w_e} = -\frac{1}{g_{\text{prot}}} \text{Re} \left[(\phi_\omega^{\text{out}})^H M^{-1} b_e b_e^T \phi_\theta^{\text{in}} \right]. \quad (35)$$

This formula explains why a line action is not judged only by flow loading or graph centrality. It is judged by the overlap between the output response path and the input forcing path through that edge.

7.3 Gate results

Gate 1 tested 60 connected Watts-Strogatz cases with $n = 5, \dots, 9$ and two IEEE topology templates, with DC flow feasibility enforced after topology changes. Costs were fixed at

$$c_w = 1, \quad c_D = 2, \quad c_M = 8.$$

Two subtests were run. Nonlinear synergy was measured by

$$\text{Syn}_\epsilon(a, b) = S(p + \epsilon a + \epsilon b) - S(p + \epsilon a) - S(p + \epsilon b) + S(p), \quad (36)$$

normalized by $|\Delta S(a)| + |\Delta S(b)| + \eta$. Equal-cost coordination was measured by

$$CV(a, b) = \Delta S_{\text{equalcost}}(a, b) - \min\{\Delta S_{\text{equalcost}}(a), \Delta S_{\text{equalcost}}(b)\}. \quad (37)$$

Negative CV means the package beats the best individual lever at the same total cost.

The nonlinear-synergy headline died. $\text{SynNorm} < -0.10$ never exceeded 50% of cases. But equal-cost coordination under S_g survived:

Gamma profile	$w + D$	$w + M$	$D + M$
identity	72.1%	67.2%	51.6%
inertia output	73.8%	73.8%	48.4%
type input low inertia	73.8%	70.5%	48.4%

Thus $w + D$ and $w + M$ are supported in this reduced-model gate; $D + M$ is not.

The certificate discrepancy also survived Gamma changes. Ranking agreement between S_ζ and S_g was 29.0% for identity, 21.0% for inertia-output, 22.6% for type-input-low-inertia, and 29.0% for the frequency-output control. Eight random positive diagonal Gamma samples did not remove the discrepancy.

The mechanism is not yet explained. The maximum absolute Spearman correlation between the tested global descriptors and ranking disagreement was only 0.216. The correct status is therefore:

Certificate discrepancy is robust to tested Gamma choices, but the global non-normality descriptors and alignment descriptors used so far do not explain it well.

7.4 Flow-feasible resolvent retrofitting algorithm

The Gemini proposal can be included as an algorithmic candidate, but only after correcting the sensitivity formulas as above and adding the status labels.

This algorithm is not yet validated on full ANDES IBR. It is the natural next step for the protected-resolvent paper.

8 Weak Nodes, Weak Links, and Substitutability

The weak-element language should be mode- and certificate-specific. A bus is not weak in the abstract. It is weak for a margin, a mode family, and a lever set.

Definition 1 (Local weak node) *For a margin S and admissible local action set $\mathcal{A}_i$ at bus i , define*

$$W_i(S) = \max_{a \in \mathcal{A}_i} [\nabla S^T a]_+. \quad (38)$$

The node is weak if its best admissible perturbation can increase the risk margin to first order.

Algorithm 1 Flow-feasible protected-resolvent retrofitting

Require: Dynamic model A , input/output matrices B, C , protection weights Γ_u, Γ_y , DC dispatch P , flow limits $\bar{f}$, action costs.

- 1: Compute g_{prot} , ω^* , and singular vectors u^*, v^* of $\Gamma_y C(j\omega I - A)^{-1} B \Gamma_u$.
- 2: Compute ϕ^{in} and ϕ^{out} .
- 3: For each bus, compute damping and inertia exchange rates using (31). For inertia, use the actual parameter map; do not substitute a virtual-inertia formula unless the controller exposes one.
- 4: For each line, compute the topology exchange rate (35) and the DC flow sensitivity/feasibility penalty.
- 5: Solve a local cost-constrained problem

$$\min_z c^T z \quad \text{s.t.} \quad S_g + \nabla S_g^T z \leq 0, \quad |f(w + \Delta w; P)| \leq \bar{f}, \quad z \in \mathcal{Z}.$$

- 6: Validate the proposed package by recomputing the full eigensolve and the protected resolvent. Reject if the local prediction is not matched.
-

Definition 2 (Repairability) *For the same node, define*

$$R_i(S) = \max_{a \in \mathcal{A}_i, c^T a \leq 1} [-\nabla S^T a]_+. \quad (39)$$

Weakness and repairability are different. A node can be exposed but expensive to repair, or easy to repair but not currently exposed.

For a line e , replace $\mathcal{A}_i$ by admissible line actions and include the DC feasibility constraint. A line reinforcement that improves a graph stiffness score but increases S_ζ or S_g is a dynamic Braess candidate. It is flow-feasible only if

$$|f(w + \Delta w; P)| \leq \bar{f}. \quad (40)$$

Substitutability compares lever subspaces. In the simplest matrix-geometric picture, existing-line changes live in

$$\mathcal{L}_{\text{existing}} = \text{span}\{b_e b_e^T : e \in E\},$$

new-line changes live in missing-edge directions, and shunt/inertia-like changes live in diagonal directions. A mode-specific substitutability score asks whether the desired pole motion can be achieved by cheap line directions or requires local equipment. This is a useful planning language. The current status is formulation plus surrogate audits, not full ANDES validation.

9 Open Gates

1. **Adversarial literature search for active nonviscous power grids.** Search terms must include active viscoelastic structures, non-dissipative nonviscous damping, active material damping kernels, and power-grid NEPs.
2. **Passivity-defect gate.** Compute (23) for no-PSS, base, Mix60, soft-GFL controls, and future calibrated IBR cases. Check whether it predicts when the scalar bridge fails.

3. **Local discrepancy mechanism.** The Gamma gate rejected global $d_F(A)$ and global χ as strong explanations. Test local descriptors: peak resolvent frequency, singular-vector spatial separation, critical-mode condition number, and local self-energy denominator fraction.
4. **Estimator validation over the NEP bridge.** Run diagonal, second-order, reduced-QEP, and full NEP/ANDES comparisons on calibrated IBR operating ranges.
5. **Planning validation.** Compare weak-node, weak-link, conversion, and coordinated-retrofitting rankings against full recomputation and against SCR, impedance, Fiedler, flow loading, damping participation, and published damping-index baselines.

10 What Is Actually New

The safest novelty statement is:

The project formulates detailed IBR small-signal damping margin analysis as a controller aware rational self-energy problem, validates the Schur NEP bridge against full ANDES eigensolves including a control-family critical mode, and uses that bridge to separate network-family and controller-family mechanisms.

The strongest current pieces are:

1. **Validated rational bridge.** The Phase-0D result is precise, non-circular, and full-ANDES-based.
2. **Control-family criticality.** Mix60's PLL-family 13.7417 Hz mode is the anchor that explains why static and fixed-point bridges fail.
3. **Resonant Braess mechanism.** Strict but minority cases show line reinforcement moving a network-family mode toward a PLL-dominated control pole.
4. **Certificate discrepancy.** S_ζ and S_g disagree under several Gamma choices; $w + D$ and $w + M$ packages have protected-resolvent coordination value. The mechanism remains open.
5. **Unified hierarchy.** The hierarchy itself is a synthesis, not a single theorem. Its value is that it prevents overclaim: each broken classical guarantee maps to a specific planning risk.

11 Submission Map

Paper 1: Controller-aware rational bridge

Core content: Level 2 bridge, Beyn contour validation, Mix60 control-family mode, resonant Braess mechanism, scalar/fixed-point failures, and negative controls. This is the nearest to a defensible submission.

Paper 2: Active nonviscous IBR grids

Core content: damping-operator hierarchy and active-material framing. This requires the passivity-defect gate and an adversarial literature search before it can be claimed.

Paper 3: Flow-feasible protected-resolvent retrofitting

Core content: bi-orthogonal resolvent sensitivity, spatial decoupling, flow-feasible retrofit LP, and Gamma-robust certificate discrepancy. This needs local mechanism descriptors and stronger comparisons against H2, SCR, impedance, Fiedler, and transmission-switching baselines.

Acknowledgment of Dead Ends

The project produced several attractive hypotheses that did not survive gates. They should remain in the internal ledger because they prevent future backsliding:

- Nonlinear three-lever synergy as a headline: killed by Gate 1.
- Universal inertia Braess: killed by fixed-ratio and Monte Carlo controls.
- Second order always beating large reduced QEP: killed by cross-checks.
- Inertia-control cross term as frequent ANDES mechanism: killed by 0/13 measurable cases.
- Virtual-inertia placement over the controller-aware bridge: blocked by missing physical GFM inertia parameter in REGF1 sheets.
- Global non-normality or global commutator defect as explanation for certificate discrepancy: weak in Gamma gate, not killed as a family but not currently supported.

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